

Machine Learning Approaches to Human Activity Recognition using Accelerometer Data

Arel Akaunu

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1 AI Disclosure

A range of AI-assisted technologies were used during the development of this coursework. These tools supported the process of refining Python code that I had written, including helping to identify and resolve bugs, optimise structure, and improve readability. Additionally, AI assistance was used to enhance the language, clarity, and formal tone of the written report.

2 Introduction

2.1 Human Activity Recognition

Human Activity Recognition (HAR) involves the automatic classification of physical movements or postures using data collected from sensors, such as accelerometers and gyroscopes. These sensors are commonly embedded in smartphones and wearable devices, enabling continuous monitoring of human behaviour in real-world settings. HAR plays a crucial role in a range of applications, including health and fitness monitoring, fall detection systems for elderly individuals, rehabilitation programmes, and human-computer interaction. Its implementation can improve quality of life, enhance safety, and contribute to public health initiatives.

The increasing availability of sensor-rich devices and the growing demand for real-time behavioural monitoring make HAR an area of significant interest. This coursework presents an opportunity to explore HAR using machine learning (ML) techniques applied to accelerometer data. The objective is to classify six human activities—walking, jogging, sitting, standing, going upstairs, and going downstairs—using a combination of signal processing, feature engineering, and supervised learning algorithms.

2.2 Overview of Approach

The methodology followed an iterative framework, beginning with an assessment of the provided data and its underlying structure through exploratory data analysis (EDA). Recognising limitations in the initial feature set, a range of additional features were engineered to improve model performance. Several classification models were then implemented and compared, including Convolutional Neural Networks (CNNs), Gradient Boosting Machines (GBMs), and Random Forests (RFs).

Model selection was based on a comprehensive evaluation of accuracy, robustness to class imbalance, and generalisability, with the final model assessed using standard classification metrics and AUROC scores. The entire workflow was informed by domain knowledge and literature in HAR, ensuring that each methodological decision was evidence-based.

3 Exploratory Data Analysis

3.1 Initial Observations

The metadata revealed that the dataset was significantly imbalanced (see [Figure 1](#)). Some classes, such as 'walking', were overrepresented, while others, particularly 'upstairs', 'downstairs', 'sitting', and 'standing' had fewer samples. This imbalance posed a risk of bias in model training, potentially leading to high overall accuracy but poor performance on minority classes.

To evaluate the separability of the activity classes using the provided features, a t-distributed stochastic neighbour embedding (t-SNE) analysis was conducted. The resulting visualisation (see [Figure 2](#)) indicated a high degree of overlap between 'walking', 'upstairs', and 'downstairs', suggesting that the existing feature set did not adequately distinguish between these activities.

Furthermore, Figure 2 shows that within each class there are clusters. This likely corresponds to different ‘styles’ of movement that are unique to smaller groups of people.

3.2 Signal Processing and Feature Engineering

After observing the overlapping of classes in the feature space from the provided metadata, new features were developed to counteract this. Alongside this, preprocessing of the signal was also carried out.

Given that the signal corresponds to accelerometer data, there is noise in the signal due to gravitational forces. It can be assumed that for a sampling rate of $20Hz$, gravitational forces will be constant. Therefore a high-pass filter was employed to remove this constant contribution from gravity. The resulting signal is now more representative of the movement of the person and not of gravity.

This filtered signal was then used to carry out feature engineering. The approach laid out by Bennasar et al. was built upon to develop 242 features. The majority of these features were in the time or frequency domains, such as Fast Fourier Transform (FFT) components and root mean square (RMS). However, some of the features included are not traditionally used in HAR such as Recurrence Quantification Analysis.

With these new features, t-SNE was carried out again with many sets of hyperparameters and the one with the clearest separation was chosen (Figure 3). The figure shows that the classes are no longer separated into further, smaller clusters which we suspect relate to unique movement signals. Furthermore, it also shows a clear separation of standing and sitting from the rest of the classes. This is expected because they are both stationary activities. Overall, the t-SNE still reveals that the classes are not well separated.

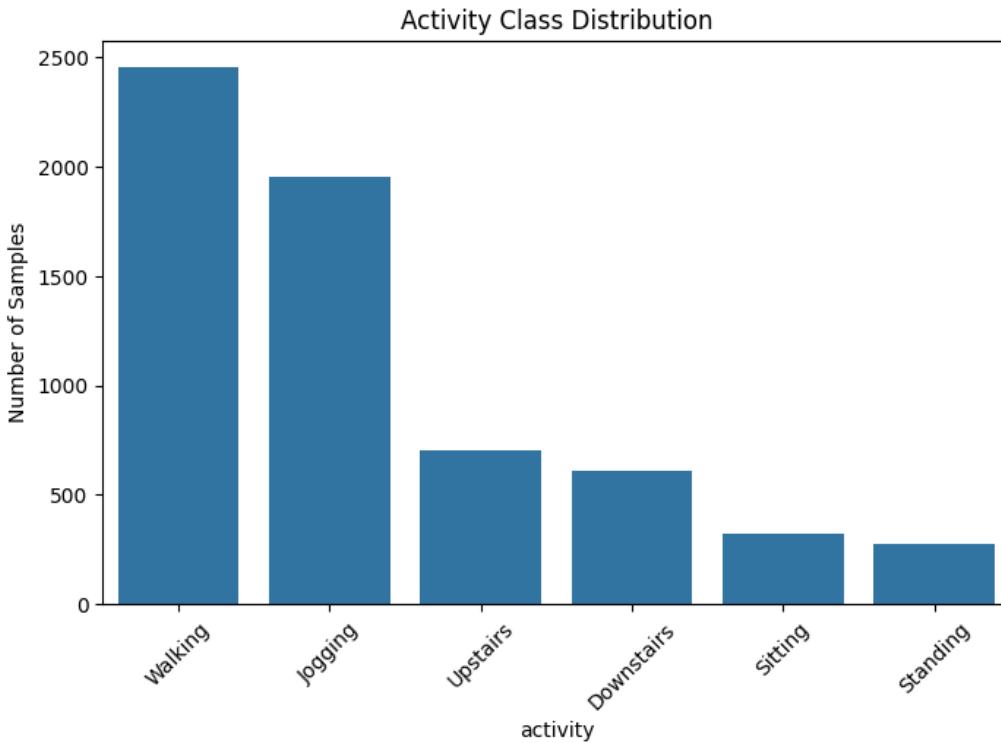


Figure 1: A bar chart of the number of samples in each class showcasing an imbalanced dataset.

4 Machine Learning Modelling

A range of supervised learning algorithms were implemented and evaluated to determine the most effective approach for the task of human activity recognition. These included a Convolutional Neural Network (CNN), Gradient Boosting Machine (GBM), and Random Forest (RF), each selected based on its suitability for structured or sequential data.

4.1 Convolutional Neural Networks

A CNN was implemented to evaluate the effectiveness of deep learning approaches for human activity recognition using the enriched feature set. Although CNNs are typically applied to raw time-series data, in this case the structured feature vectors were reshaped to simulate a one-dimensional signal input.



Figure 2: A t-SNE plot of the features provided in the metadata

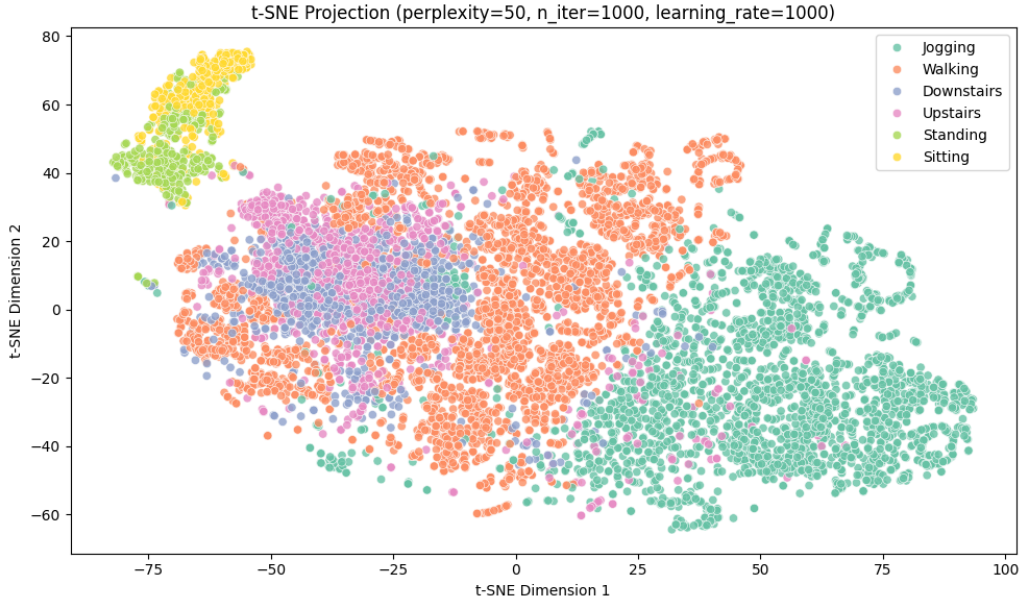


Figure 3: A t-SNE plot of the new features developed (Bennasar et al., 2022).

The architecture consisted of five consecutive 1D convolutional layers, each with 16 filters and a kernel size of 8, interspersed with dropout layers to reduce overfitting. A final Flatten layer and Dense output layer with softmax activation were used for multiclass classification across the six activity labels.

Before training, the target variable was one-hot encoded to ensure compatibility with the categorical crossentropy loss function and softmax output. The model was compiled and trained using the Adam optimiser over 200 epochs with a batch size of 64 and a validation split of 20%.

Despite its deep architecture and regularisation, the CNN achieved a test accuracy of approximately 81%. While this result was acceptable, it underperformed compared to the GBM and RF models, especially on minority classes. Additionally, the model required significantly more training time and computational resources. Due to these limitations and its lower generalisability, the CNN was not selected for final deployment.

In future a deep CNN should be employed on the raw time series data in order to gain insights without feature engineering. However, this may require extensive computational power.

4.2 Gradient Boosting Machine

GBMs were selected as a core modelling approach due to their strong performance with structured, high-dimensional data and robustness to class imbalance. GBMs allow the model to capture complex non-linear interactions without requiring extensive preprocessing.

Initially, an XGBoost classifier was used to perform feature importance ranking on the combined metadata and frequency-domain feature set. Based on this ranking, the 100 least important features were removed in an attempt to simplify the model and improve generalisability by reducing the risk of overfitting. However, when this reduced feature set was used to train a GBM, a decline in classification accuracy was observed. The model achieved slightly lower performance across both training and validation sets compared to using the full feature set. As a result, all features were retained in the final GBM training pipeline.

The final GBM was implemented using scikit-learn’s GradientBoostingClassifier. Missing values in the dataset were imputed using the mean of each column. No feature scaling was required, as it is not required for decision tree-based methods. The model was trained using default hyperparameters, which already yielded high performance. A grid search was later conducted to explore tuning options such as the number of estimators, learning rate, and maximum depth, but no significant improvement in accuracy was observed. Given the diminishing returns from tuning, the default configuration was retained for its simplicity and efficiency.

The final model achieved an accuracy of 90% on the test set and demonstrated consistent performance across both majority and minority classes (see Figure 4). Notably, the GBM achieved this without the use of oversampling techniques such as SMOTE, further confirming its robustness to imbalance. Feature importance outputs from the model confirmed that both statistical time-domain features and engineered frequency-domain components contributed meaningfully to the model’s decision process.

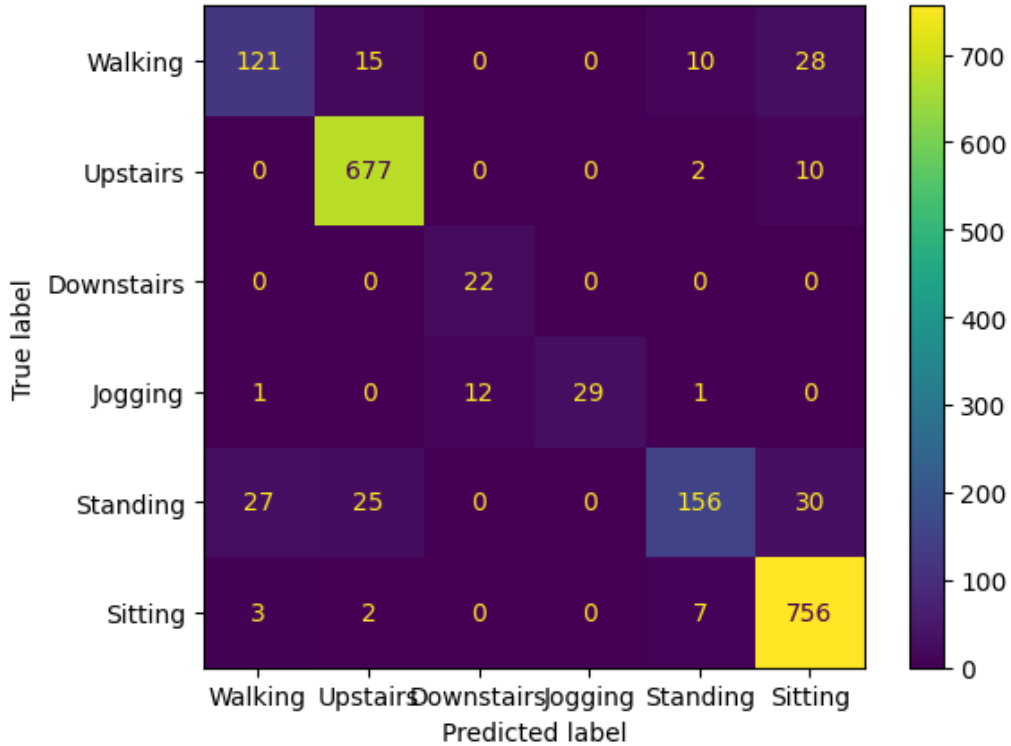


Figure 4: A confusion matrix showing the performance of the Gradient Boosting Machine model on the test set.

4.3 Random Forest

Random Forest was implemented as a comparative tree-based ensemble model to benchmark performance against the Gradient Boosting Machine. Initially, a baseline model was trained with a standard configuration using 100 estimators. To address the notable class imbalance in the dataset, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. This method generated synthetic examples of minority classes to ensure a more balanced class distribution and reduce the likelihood of biased predictions.

Following the baseline evaluation, a systematic grid search was carried out to optimise the model’s hyperparameters. Stratified five-fold cross-validation was employed to ensure robustness across all activity classes.

The evaluation metric used was the F1-weighted score, chosen for its ability to reflect both precision and recall while accounting for class imbalance. The grid search explored a range of parameters, including the number of estimators, tree depth, and constraints on node splitting.

The best-performing Random Forest model achieved an accuracy of 89% on the test set. However, it frequently misclassified 'upstairs' (see Figure 5). While this result was comparable to the GBM, it required additional preprocessing in the form of oversampling and extensive parameter tuning. Furthermore, as discussed later in the report, signs of overfitting were observed in the model's AUROC scores, particularly on minority classes. For these reasons, the Random Forest model, despite its competitive accuracy, was not selected as the final solution.

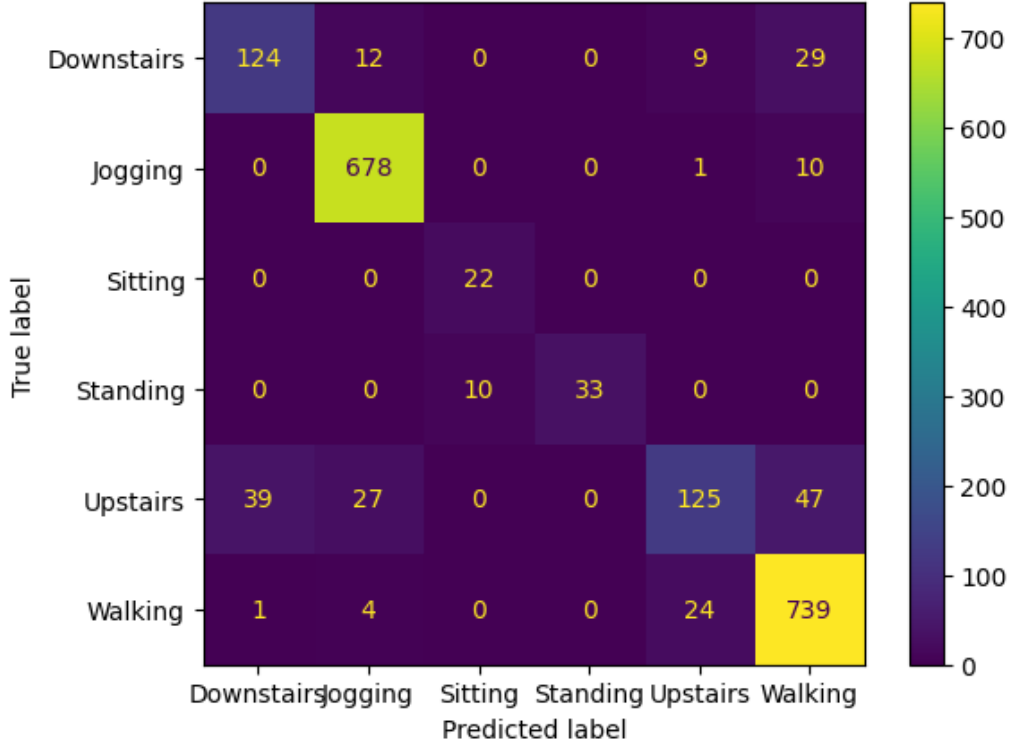


Figure 5: A confusion matrix showing the performance of the Random Forest model on the test set.

5 Discussion

The task of classifying human activities using sensor-derived features presented several challenges, most notably class imbalance and overlap in feature space between physically similar actions. The initial exploratory analysis highlighted that activities such as walking, upstairs, and downstairs shared similar statistical properties. This was confirmed through t-SNE visualisations, which revealed a lack of clear separation between these classes when using the provided metadata. The application of high-pass filtering, followed by frequency-domain feature engineering, modestly improved class separability, though some overlap remained, suggesting a fundamental similarity in the recorded acceleration patterns of those movements.

Three machine learning models were evaluated: CNN, GBM, and RF. The CNN, despite being a suitable candidate for temporal data, struggled in this case due to the structured tabular nature of the input and the relatively small number of samples per class. Although the architecture included regularisation via dropout and training was extended to 200 epochs, the model reached a plateau at 81% accuracy and exhibited reduced performance on underrepresented classes. The need for one-hot encoding and significant computational time further reduced its practicality. In retrospect, a larger dataset with raw time-series input may have better supported a CNN-based approach.

In contrast, the GBM demonstrated excellent performance, achieving 90% accuracy with minimal tuning and no need for class rebalancing. This outcome reflects the model's strength in handling structured data and learning from imbalanced datasets without extensive intervention. AUROC analysis also showed that the model performed well (see Figure 6). Attempts to improve generalisability through feature selection, specifically by removing the bottom 100 features ranked by an XGBoost importance score, led to a slight drop in accuracy. This indicates that even seemingly low-importance features contributed useful information when considered as

part of the ensemble. Thus, the decision to retain all engineered features was justified.

The Random Forest model also performed strongly, reaching 89% accuracy after tuning via grid search and applying SMOTE to balance the training set. However, unlike GBM, it required more extensive preprocessing and was more sensitive to class imbalance. Furthermore, AUROC analysis revealed unusually high class-wise scores for RF, including perfect AUCs for several classes (see [Figure 7](#)). This was indicative of overfitting to the oversampled training data and reinforced the decision not to proceed with it as the final model. By comparison, the GBM produced more modest but realistic AUROC scores, indicating stronger generalisation.

The confusion matrix further supported these findings. GBM handled all classes consistently, with very few misclassifications, though some confusion persisted between the similar locomotive activities. This was expected, as such classes are inherently difficult to separate using tri-axial acceleration data alone, especially at a sampling rate of 20Hz. Nonetheless, GBM’s ability to correctly classify examples from both dominant and minority classes without external balancing techniques underscores its robustness and suitability for this type of task.

5.1 Future Improvements

While the final results were strong, several improvements could be made to enhance model performance and reliability. First, sourcing or collecting a more balanced dataset would likely improve generalisability, particularly for underrepresented activities. Second, increasing the sampling rate would capture more nuanced aspects of motion, potentially allowing models—particularly deep learning approaches—to distinguish between similar classes more effectively. In addition, integrating gyroscope data alongside accelerometer inputs could introduce rotational information that may help differentiate between activities like stair climbing and level walking.

Another method that might increase accuracy is to use a smaller sliding window. This would let the engineered features capture details at higher frequencies which may be important in distinguishing high movement activities.

From a modelling perspective, experimenting with hybrid architectures, such as CNN-LSTM models, may allow for more effective use of temporal patterns in raw time-series data. These approaches could be combined with handcrafted features in an ensemble or stacked framework to improve both interpretability and performance. Finally, employing techniques such as model calibration or uncertainty estimation would help quantify confidence in predictions, an essential step for real-world deployment in safety-critical applications like fall detection.

6 Conclusions

The project demonstrated a rigorous and methodical approach to solving a real-world classification problem using machine learning. Key strengths included a sound exploratory analysis, thoughtful feature engineering, and critical model selection based on both empirical results and practical considerations. While there were some limitations—particularly related to data resolution and class overlap—these were recognised early and the class overlap and imbalance were addressed with appropriate techniques. The final GBM model achieved a high level of accuracy and generalisability, while the iterative experimentation process provided valuable insights into the strengths and limitations of different modelling strategies. Future iterations would benefit from improved data and the inclusion of temporal modelling methods, but the current outcome represents a solid and effective solution to the problem.

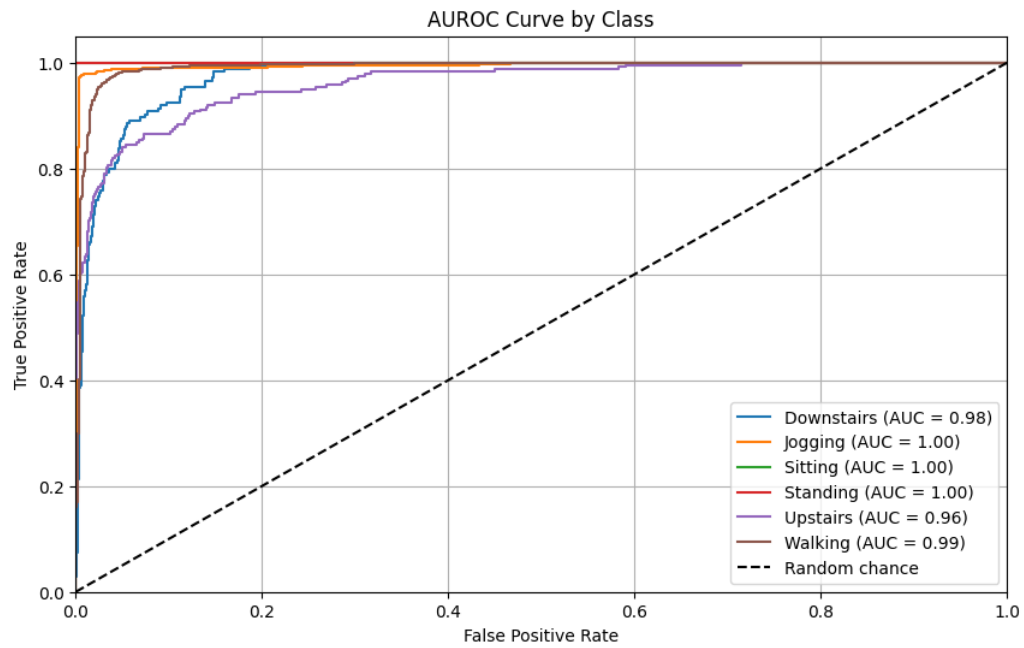


Figure 6: AUROC for the Gradient Boosting Machine Model.

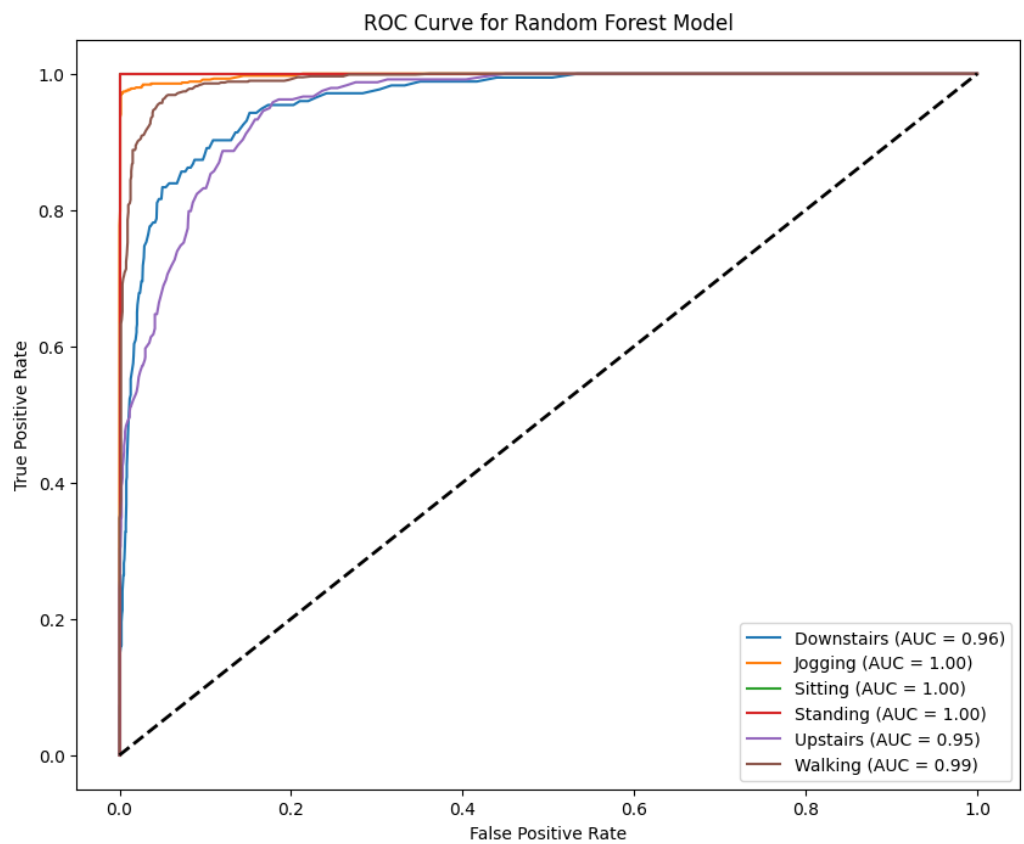


Figure 7: AUROC for the Gradient Boosting Machine Model.

References

Bennasar, M. et al. (2022). “Significant Features for Human Activity Recognition Using Tri-Axial Accelerometers”. In: *Sensors* 22.19. Number: 19 Publisher: Multidisciplinary Digital Publishing Institute, p. 7482. ISSN: 1424-8220. DOI: [10.3390/s22197482](https://doi.org/10.3390/s22197482). URL: <https://www.mdpi.com/1424-8220/22/19/7482> Accessed 04/04/2025.